

Tracking

One of the most important features of a GPS system is its ability to let you lay out a track. A track is like a path that you create which tells you how you have reached a particular place. This differs from a "route," which details where you're going. You can configure a GPS to automatically drop "trackpoints" over intervals of either time or distance. To retrace your steps, simply follow the GPS bearings back through the sequence of trackpoints.

To provide reliable navigational information a GPS receiver needs to receive good signals from at least four satellites.

In order to acquire the signals from the satellites, all that needs to be done is to turn on the GPS receiver and go to the satellite screen. This will display the current configuration of the satellites and the strength of the signals. It may take several minutes for the GPS unit to lock in to the satellites. In case there are only a few satellites present or the signals from the satellites are all weak, it is always advisable to simply use a map and a compass.

Most GPS receivers automatically select the satellites used in the timing solution. This makes them easy to use. Often, you simply turn the receiver on and wait for a signal to be acquired. However, the algorithms used by all the GPS receivers are all different. The decision to keep or to drop a particular satellite's signal is also completely based on the algorithm. Some algorithms choose the satellites that provide the best geometric dilution of precision (GDOP). Others choose the satellites highest in the sky after a fixed position has been entered. Some algorithms limit the timing solution to just one, or just a few satellites. Others can use as many as 12 satellites in the solution. For this reason, two GPS receivers can obtain very different results even when connected to the same antenna in the same location.

The optimum satellite lock occurs when there is a clear view of the sky above. Tree canopy, canyons and tall buildings that obscure the view overhead or obscure the view of the horizon

Depending on the algorithm used by the GPS system, two receivers mounted even on the same antenna can give varied results